TEST PLAN sample

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5. Introduction

This document describes the accreditation requirements for farmers' assistant application. This program is designed to meet the needs for animal feeds system in order to facilitate accountability, efficiency and customer satisfaction.

Testing is a major source of feedback and serves as a benchmark whether the system is fool proof. Testing software will be done to ensure the system produces desired results are achieved under unknown conditions. This process generates a high-level test plan document that identifies the software items to be tested, the degree of tester independence, the test environment, the test case design and test measurement techniques to be used and rationale of their choice.

The system will do the following:

i. Print various results.
ii. Handle user sessions
iii. Provide the users with menus, directions and error messages to direct them on various options.
iv. Process database values
v. Handle update, addition and delete queries
vi. Handle search queries.
vii. Show alerts when stock is running low
viii. Generate reports for financial periods
ix. Coordinate chat queries.

Its objective is to communicate project-wide quality standards and procedures. It portrays a snapshot of the project as of the end of the planning phase. This document will address the different standards that will to unit integration and system testing of the specified application. Testing criteria will be utilized under the white box, black box and system testing paradigm. This paradigm will include, but is not limited to the testing the testing criteria, method and test cases of the overall design.

The scope of accreditation is specific and users are encouraged to ask to see the scope of accreditation to review those specific before employing the farmers' assistant application.

5.1 Goals and Objective

The main goal of testing is to come up with a detailed document that will be used by the developer and the user to satisfy that the system is able to perform as per the system requirements.

i. The system test should help identify errors, faults, bugs and failures in the system so that it can be described as a successful test.
ii. It should help to establish test cases for the system and also test data to enable an efficient and effective system test.
iii. It should help to establish resources for performing the test in the test plan. These resources include money required for manpower.
iv. The test schedule is also developed for performing the tests which allocates the time required for each part of testing and the procedures to be followed.
v. Ensuring that it meets all user requirements as stated in the system requirement specification.
vi. The developed software should be easy to use and understand to all personnel in the organization.
vii. The system should also ensure consistency in the software that has been developed ensuring there are no deviations from the specifications.

5.2 Statement of scope

The primary purpose for testing is to detect software failures so that defects may be uncovered and corrected as early as possible during the development stage. This is a non-trivial pursuit.

Testing cannot establish that a product functions properly under all conditions but can only establish that it does not function properly under specific conditions. The scope of this system testing will include examination of code as well as the execution of that code in various environments and conditions as well as examining the aspects of code: does it do

what it is supposed to do and also do what it is needed to do. In case the client requests, a testing organization may be invited to test the system.

Information derived from the system testing will be used to correct the process by which the software is developed.

The very fundamental area to be tested in the system include inputs, outputs, functional requirements and non-functional dimensions of quality (how it is supposed to be versus what it is supposed to do). For example; usability, scalability, performance, compatibility, security, maintainability, reliability – can be highly subjective; something that constitutes sufficient value to one person may be intolerable to another.

5.2.1 Major constraints

The system can run in one computer and fail to run in another computer. This could be due to difference in server software, the operating system in use.

Also, the availability of the necessary software in all the computers to be used. For example, a computer may not have XAMPP server so in this case you would have to install it.

5.2.1.1 software to be tested

The software to be tested: farmers' assistant system to prove whether it provides the identified objectives.

5.2.1.2 Testing strategy

The test strategy consists of various test that will ensure the system is tested appropriately in units and as a whole. This will assist uncover errors in the system in the unit modules and during their integration and as a whole. It outlines an ordering for the steps in the overall testing process.

5.2.1.3 Unit testing

Unit testing is where individual components are tested independently to ensure they are operating correctly before being integrated.

Unit testing is normally the first test that is completed during the configuration effort and is focused towards the program's inner functions, rather than towards the integration.

Type of test	**Area being tested**	**Expected output**	**Actual outcome**	**Incorrect outcome.**
Unit test	**inputs**			
	Password	It should not show password characters.	Output result of the password in the screen ****	The user cannot evade authentication is redirected to login page.
	Username	Usernames registered can be used	Being logged in	Error message and redirect to login
	User category	User should select radio button for their various categories	Either admin, customer or doctor	Error message and redirect to login
	Mpesa code	Payment code is mandatory for online payment.	Code should be valid and is to be printed in receipt.	Error message incorrect or blank code.
	Numeric fields	It will accept numerical values only.	Proceed to the next field entry	Error message incorrect format.

	Date	It will accept date being in the right format	Proceed to the next entry.	Error message incorrect date.
	Alphabetic fields	This field will accept alphanumerical characters	Proceed to the next entry.	Error message invalid
	Indexed fields	They cannot be left blank	Proceed to the next entry.	Error message incorrect format.
Screen design and layout.	**Layout**	Ease of use of interfaces.	All interfaces of the system are in a format that the user can easily find information they need.	
Output design components	**Output**	Ease of producing reports in a clear format.	The reports should be easily produced as they are sort.	
Database	**Storage**	Correct storing and retrieval of information in the database.	The information entered should be stored in a correct format.	

5.2.2 Integration testing

A system comprises of various different modules which have been developed separately and have to be combined to come up with a fully functional system that is ready for delivery to the client. After the modules have been fully tested multiple components are combined.

Integration testing is accomplished incrementally by adding one module at a time to isolate errors. Integration testing is systematic technique for constructing the program structure while at the same time conducting test to uncover errors associated with interfacing. The objective is to take unit tested modules and build a program structure as directed by the design.

It is also concerned with validating that the system meets its functional and non-functional requirements.

Functional requirements include:

i. A functional login and signup mechanism
ii. Consistency and integrity of the database
iii. Accuracy of data entry
iv. Accurate output from the system.
v. Effective information search mechanism
vi. Report generation
vii. Usability in terms of system interface requirements.

Type of test	**area being tested**	**Expected outcome**	**Actual outcome**
Integration test	relationship	Entities that are logically related should be able to share data.	Defines relationship between entities.
	reports	Reports should be produced for each transaction	Generates relevant reports for specific users.

5.2.3 validation testing

Validation testing is sometimes called high order testing, tests the software to validate that it functions in a manner that the end user would expect.

This test gives two possible outcomes: the function or performance characteristics conforms to specifications and are accepted and a deviation from the specification is uncovered and a deficiency list is uncovered.

type of testing	**Area being tested**	**Expected outcome**	**Actual outcome**
Validation testing	**Forms**		
	Add module	ease of adding records to the system	insert new records to the database.
	delete module	Ease of removing existing records from the database	Delete records from the database.
	Menu module	Should open the required area	The area being displayed
	update module	Details to be displayed if it exists	The current details are being updated
	save module	Should save existing and new entries to the database	The data being solved
	Display module	Ability to display database information when required	Display the content of a table in the database.
	Search module	Ability to find information from the storage	Retrieve a record from the database

	Sort module	Ability to manipulate storage order	Arrange records as specified.
	Previous module	Ability to move the previous record	Move to the previous record
	Next module	Ability to move to the next record in the database	Move to the

5.2.4 High order testing (system testing)

This involves carrying out a test after the different components for the entire system have been integrated. This process is aimed at finding errors that may arise as the different components interact with each other. It also ensures that the whole system meets its functional requirements and non-functional requirements.

Test component	**status**	**recommendation**
Login form	pass	Validate users.
Signup form	pass	Create users
Chat form	pass	Communication between doctor and customer.
Add products form	pass	Add products to various categories
Add customer information form	pass	Add customers prescription details
Checkout form	pass	Buy products from the online platform.
Point of sale form	pass	Allow cashier sell at the counter.

Report form	pass	Show transaction details for specified time range
Receipt form	pass	Printing receipt after each sale.
Payment form	pass	Enable enter payment details

5.2.4.1 recovery testing

Recovery testing is the activity of testing how well the application is able to recover from crashes, hardware failures and other similar problems.

Recovery testing is the forced failure of the system in a variety of ways to verify that recovery is properly performed.

Some of the recovery testing that will be used include:

i. Switching off server when the application is loading data after that checking the validity of applications data integrity.
ii. Sudden restarting the computer while the system is running and after that checking the validity of applications data integrity.
iii. Unplug then plugin data cable when the application is getting data from an external source and observe if it continues.
iv. Redirect URL to observe if the key authentication forms can be bypassed by the user session.

5.2.4.2 security testing

This is the process to determine that the system protects data and maintains its functionality as intended.

Concepts covered in security testing are: confidentiality, integrity, authentication, authorization, availability and non-repudiation.

i. Confidentiality – a security measure which protects against the disclosure of information to parties other than the intended recipient.
ii. Integrity – a measure intended to allow the receiver to determine that the information which it receives has not been altered in transit or by other than the originator of the information.
iii. Authorization – process of determining that the requestor is allowed to receive a service or perform an operation. This will ensure access control to the system.

iv. Availability – this entails assuring information and services will be ready for use when expected. It will also ensure that information is kept available to authorized persons when they need it.

5.2.4.3 performance testing

Performance testing is testing performed from one perspective, to determine how fast some aspect of the system performs under a particular workload. It will also serve to validate and verify other quality attributes of the system such as scalability, reliability and resource usage.

Performance testing can used to determine if it meets the performance criteria. It can compare two systems to find which performs better. It can measure what parts of the system or workloads that cause the system to perform badly. In the diagnostic case, software engineers use tools such as profilers to measure what parts of a device or software contribute most to poor performance or to establish throughput levels for maintained acceptable response time.

The later a performance defect is detected, the higher the cost of remediation.

Purposes

i. Demonstrate that the system meets performance criteria.
ii. Compare two systems to find which performs better.
iii. Measure what parts of the system or workload cause the system to perform badly.

5.2.4.4 Alpha/Beta testing

Before compiling the final system version, alpha and beta testing will be contracted.

Alpha testing is simulated or actual operational testing by potential users/customers or independent test team at the developers' test.

This ensures that the system is tested against a complete set of requirements. Users will get the system and do their own testing and report findings to the developer (GRAHAM) before a final release can be released.

5.3 testing resource and staffing

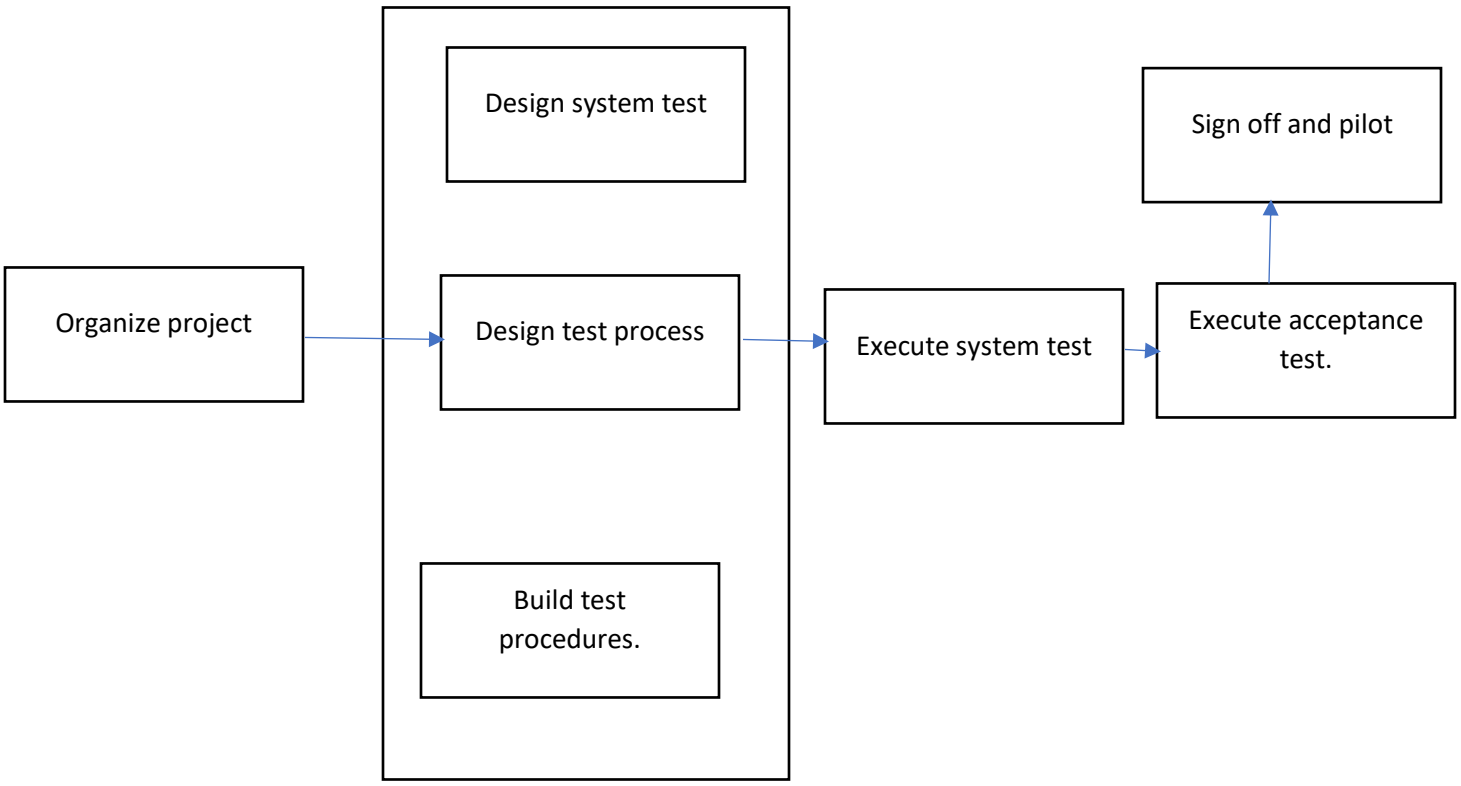
The testing process would require a computer installed with necessary software, farmers administrator, farmers cashier, farmers advisor and a customer. All the above persons would participate in the testing process.

Users are trained on how to use the system and let them use the system to ensure that everything is as anticipated.

Type of test	**Areas of interest**	**Expected results**	**Actual results**

Acceptance testing			
	Training	Users are trained on how to use the new system.	Users are able to learn quickly.
	User manual	Gives a detailed explanation of the whole system.	Users understand it quickly.

5.4 Testing procedure



Organize the project – this involves creating a system test plan, schedules and test approach as well as assigning resources.

Design/build system test – involves identifying the test cycles, test cases, entrance and exit criteria, expected results. Expected results are identified by the test team.

Design/ build test procedures – includes setting up procedures such as error management systems and status reporting and setting up the data tables for the automated testing tool.

Build test environment – it entails requesting / building hardware, software and data setups.

Execute project integration test

Execute operations acceptance test

Sign off – happens when all predefined exit criteria have been achieved.

5.6 test record keeping

A detailed test report from the test team should be prepared, documented and bound for use for design and future reference.